FEASIFY FUNCTIONAL TEST CASES & EXECUTION MATRIX

Document ID: FEASIFY-DOC-003
Version: 1.0.0
Date: November 10, 2026
Author: Frontend & Backend Functional Testing Agents
Classification: Quality Assurance Test Execution Matrix

1. Execution Summary

This document presents the detailed functional test case matrix executed against the Feasify application. Test statuses are classified as **Passed**, **Failed**, **Blocked**, or **Not Executed**.

- **Total Planned Cases:** 42
- **Executed Cases:** 42
- **Passed:** 37
- **Failed:** 5
- **Blocked:** 0
- **Pass Rate: 88.1%**

2. Test Case Execution Matrix

TEST CASE ID	MODULE	TEST SCENARIO	TEST STEPS	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-AUTH-001	Authentication	Valid Student Login	1. Navigate to `/` 2. Enter student email and password 3. Click "Sign In"	User authenticated; redirected to `/dashboard`	Authenticated successfully; redirected to `/dashboard`	**Passed**
TC-AUTH-002	Authentication	Invalid Password Handling	1. Navigate to `/` 2. Enter registered email with invalid password 3. Click "Sign In"	Error message displayed: "Invalid credentials"	Error alert modal displayed; login prevented	**Passed**
TC-AUTH-003	Authentication	Password Recovery Dispatch	1. Click "Forgot Password" 2. Enter email 3. Click "Send Reset Link"	Password reset email dispatched via Firebase Auth	Firebase reset email dispatched; success banner shown	**Passed**
TC-AUTH-004	Authentication	Hardcoded Chairperson Account Privileges	1. Login with `chairperson@gmail.com` 2. Observe assigned role	User strictly assigned Chairperson role regardless of DB record	Email `chairperson@gmail.com` bypasses DB role lookup and assigns Chairperson role	**Passed (Design Concern)**
TC-AUTH-005	Authentication	Session Expiration & Logout	1. Click user avatar 2. Click "Sign Out" 3. Attempt to access `/projects`	Session cleared; redirected to `/`	User signed out; route protected; redirected to `/`	**Passed**
TC-RBAC-001	Access Control	Student Route Protection	1. Log in as Student 2. Manually navigate to `/admin/users`	Access denied; redirected to `/dashboard`	`ProtectedRoute.tsx` blocks unauthorized access and redirects	**Passed**
TC-RBAC-002	Access Control	Adviser Route Protection	1. Log in as Adviser 2. Manually navigate to `/admin/settings`	Access denied; redirected to `/adviser/dashboard`	Route guard blocks access and redirects to adviser dashboard	**Passed**
TC-RBAC-003	Access Control	Student Direct Profile Manipulation	1. Student opens Settings 2. Attempts to update username handle	Username updated and audit logged	**Failed:** Throws `ReferenceError: newUsername is not defined` (FEASIFY-BUG-002)	**Failed**
TC-PROP-001	Proposal Management	0 Proposals Baseline	1. New group logs into `/projects` 2. Query proposal count	Count is 0; "Create Proposal" available	Count is 0; creation modal fully accessible	**Passed**
TC-PROP-002	Proposal Management	1st Proposal Creation	1. Fill basic proposal form 2. Click "Save as Draft"	Proposal saved with proposalNumber = 1	Saved to Firestore; proposalNumber 1 assigned	**Passed**
TC-PROP-003	Proposal Management	2nd Proposal Creation	1. Fill second proposal details 2. Click "Save as Draft"	Proposal saved with proposalNumber = 2	Saved to Firestore; proposalNumber 2 assigned	**Passed**
TC-PROP-004	Proposal Management	3rd Proposal Creation (Limit Ceiling)	1. Fill third proposal details 2. Click "Submit Proposal"	Proposal saved with proposalNumber = 3; limit reached	Saved to Firestore; proposals array length reaches 3	**Passed**
TC-PROP-005	Proposal Management	4th Proposal Creation Attempt (Client)	1. Team with 3 proposals attempts to open creation form	System blocks creation; Toast displays "Proposal Limit Reached"	Blocked by `proposals.length ≥ 3` check; toast displayed	**Passed**
TC-PROP-006	Proposal Management	4th Proposal Creation Attempt (Server API)	1. Send `POST /api/teams/:groupId/proposals` when group has 3 proposals	Server responds with HTTP 400 and `proposalLimitReached: true`	Responded with HTTP 400 and `proposalLimitReached: true`	**Passed**
TC-PROP-007	Proposal Management	Browser Refresh at 3-Proposal Limit	1. Refresh browser when 3 proposals exist 2. Check button states	Proposal list renders; "Create New" disabled; proposals intact	`sessionStorage` and Firestore snapshot restore state; creation blocked	**Passed**
TC-PROP-	Proposal Management	Edit Existing Proposal at 3-	1. Select Proposal 2 from list 2. Update tagline and marketing details	Update permitted; count remains 3	`updateDoc` executed successfully; proposal updated	**Passed**

TEST CASE ID	MODULE	TEST SCENARIO	TEST STEPS	EXPECTED RESULT	ACTUAL RESULT	STATUS
008		Proposal Limit	3. Click "Save Proposal"			
TC-PROP-009	Proposal Management	Direct Firestore Client Injection Bypass	1. Invoke `addDoc(collection(db, "proposals"))` directly via console	Operation rejected by backend/database rule	**Failed:** Allowed! Missing Firestore database-level quota security rule (FEASIFY-BUG-003)	**Failed**
TC-PROP-010	Proposal Management	Simultaneous Concurrent Submissions (Race Condition)	1. Two students in same group click submit simultaneously at 2 proposals	Exactly one succeeds; 4th proposal prevented	Vulnerable to TOCTOU race condition without database transaction lock	**Failed (High Risk)**
TC-BTN-001	Submit Button	Form Field Validation on Submit	1. Leave required fields empty 2. Click "Submit Proposal"	Validation error modal highlights missing fields	`submissionFailureReasons` modal displays missing fields	**Passed**
TC-BTN-002	Submit Button	Negative Capital Validation	1. Enter `totalCapital: -50000` 2. Click "Submit Proposal"	Submission rejected with error: "Capital cannot be negative"	`checkTotalCapital` flagged error; submission blocked	**Passed**
TC-BTN-003	Submit Button	Button Loading State & Idempotency	1. Click "Submit Proposal" 2. Rapidly double-click button	Button disabled; spinner rendered; duplicate submission prevented	`isSubmitting` state sets button to disabled with `<Loader2>`	**Passed**
TC-BTN-004	Submit Button	Submission Success Feedback	1. Submit valid proposal 2. Observe UI	Green success notification toast appears; status updates	Status badge updates to "Submitted"; toast displayed	**Passed**
TC-FIN-001	Financial Input	Product Costing Calculation	1. Add product: 40 units, 4 batches/mo 2. Enter ingredients totaling ₱1,000 3. Set 50% markup	Selling price = ₱37.50; Monthly Revenue = ₱6,000	Math engine calculates exact unit COGS and revenue	**Passed**
TC-FIN-002	Financial Input	Operating Expenses (OPEX) Normalization	1. Add Rent (₱15,000) and Utilities (₱8,000) 2. Inspect total OPEX	Total monthly OPEX = ₱23,000; Annual OPEX = ₱276,000	`normalizeOpexList` aggregates monthly and annual OPEX	**Passed**
TC-FIN-003	Financial Input	Five-Year Balance Sheet Balance Invariant	1. Enter capital of ₱200,000 2. Generate 5-year statement	Assets == Liabilities + Equity across all 5 years	Equation holds with zero mathematical variance	**Passed**
TC-FIN-004	Financial Input	Severe Deficit Handling (CapEx > Capital)	1. Enter CapEx of ₱500,000 with initial capital of ₱100,000	System computes bank financing / notes payable liability	Balance sheet reflects liability offset; equation balances	**Passed**
TC-FIN-005	Financial Input	Philippine BMBE vs Regular Tax	1. Select BMBE tax tier 2. Inspect income tax line	Income tax computed as 0% (BMBE exempt under RA 9178)	Income tax reflects ₱0 with 3% percentage tax handled	**Passed**
TC-NAME-001	Name Checker	Empty Query Handling	1. Send empty string to name checker	Responds with error status and prompt to enter name	Status `ERROR` returned with validation message	**Passed**
TC-NAME-002	Name Checker	Maximum Length Boundary (>150 chars)	1. Query name with 151 characters	Responds with error: exceeds 150 characters	Rejected with `ERROR`: "query exceeds 150 characters"	**Passed**
TC-NAME-003	Name Checker	Whitespace Normalization	1. Query name with extra spaces: `" cocopan bakery "`	Query normalized to single spaces	Normalization trims edges, but preserves internal spaces in `buildResult` (FEASIFY-BUG-004)	**Failed (Minor)**
TC-NAME-	Name Checker	Trademark Collision Alert	1. Query "Goldilocks Bakeshop"	System flags trademark collision alert	Trademark collision alert returned with brand "goldilocks"	**Passed**

TEST CASE ID	MODULE	TEST SCENARIO	TEST STEPS	EXPECTED RESULT	ACTUAL RESULT	STATUS
004						
TC-NAME-005	Name Checker	Nonexistent Business Name Result	1. Query unique original name	System returns `NOT_FOUND` with official BNRS link	Returns `NOT_FOUND`; does NOT claim "officially available"	**Passed**
TC-NAME-006	Name Checker	SEC eSPARC Reservation Distinction	1. Query corporate name in SEC provider	Disclaimer states lookup is not official reservation	Explicit disclaimer shown directing to SEC eSPARC	**Passed**
TC-MAP-001	Location Map	Leaflet Map Initialization	1. Navigate to `/projects` 2. Open location picker	Leaflet map container initializes with OSM tile layer	Map renders correctly with custom red pin	**Passed**
TC-MAP-002	Location Map	Interactive Marker Drag / Click	1. Click map location in Valenzuela City	Pin moves; reverse geocoding queries Nominatim	Pin moves; clean address formatted without raw coordinates	**Passed**
TC-MAP-003	Location Map	Philippine Landmark Search	1. Type "PLV" or "SM City Valenzuela"	Instant suggestions populated from local catalog	Autocomplete list displays landmark with clean coordinates	**Passed**
TC-MAP-004	Location Map	Address Persistence on Proposal Reopen	1. Save proposal with selected address 2. Reopen proposal	Saved address displayed identically	Address string restored from Firestore proposal document	**Passed**
TC-ADV-001	Adviser Workflow	Assigned Section Proposal Visibility	1. Log in as Adviser 2. View section proposals	Only proposals matching adviser's sections displayed	Proposals filtered strictly by adviser's assigned sections	**Passed**
TC-ADV-002	Adviser Workflow	Proposal Evaluation & Remarks Submission	1. Review proposal 2. Add remarks 3. Click "Request Revision"	Status updates to "Revision Required"; remarks saved	Firestore document updated; notification sent to student	**Passed**
TC-ADV-003	Adviser Workflow	Proposal Status Mutation Auth Bypass	1. Send `PUT /api/proposals/:id/status` without Auth token	Request rejected with HTTP 401 Unauthorized	**Failed:** Request allowed! Missing authentication on backend endpoint (FEASIFY-BUG-005)	**Failed (Critical)**
TC-CHAIR-001	Chairperson	Batch Student Import via Excel	1. Upload student roster `.xlsx` 2. Process import	Accounts created in secondary Auth; profiles saved to Firestore	Students created and email credentials dispatched	**Passed**
TC-CHAIR-002	Chairperson	Plaintext Password Storage in Firestore	1. Inspect created user document in `users` collection	Password must NOT be stored in plaintext	**Failed:** Plaintext password stored in `users/{uid}.password` (FEASIFY-BUG-006)	**Failed (Critical)**